

Reproducing and Experimentation on Multimodal mRedditSum

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Abstract

Multimodal discussion threads on social media platforms combine textual opinions with visual context, making automated summarization a challenging task. MRedditSum introduced a large-scale dataset of Reddit threads paired with images and proposed a Cluster-based Multi-Stage Summarization (CMS) pipeline to address this problem. In this work, we reproduce the CMS approach using T5-based models and evaluate its performance on the MRedditSum dataset. We conduct controlled experiments comparing text-only and image-caption-based inputs to quantify the contribution of visual information. In addition, we explore semantic comment clustering as a preprocessing strategy to structure long discussion threads prior to summarization. Our evaluation uses ROUGE metrics across standard train, validation, and test splits, and includes qualitative case studies. This work serves as both a replication study and an extension of prior multimodal discussion summarization research.

1 Introduction

Online discussion forums such as Reddit host large volumes of user-generated content where individuals seek advice, share opinions, and discuss real-world problems. These discussions often involve long comment threads containing diverse and sometimes redundant viewpoints, making it difficult for users to quickly extract useful information. Automated summarization has been proposed as a solution to this problem, but most prior work focuses on text-only data.

In practice, many discussion threads are multimodal, combining textual content with images that provide essential context. For example, posts seeking advice on interior design, fashion, or home improvement often rely heavily on visual information. Summaries that ignore this visual context risk missing critical details or producing incomplete interpretations.

MRedditSum addresses this gap by introducing a multimodal dataset of Reddit threads paired with images and human-written summaries. The authors also propose CMS, a Cluster-based Multi-Stage Summarization framework that first groups comments by semantic similarity and then synthesizes summaries in multiple stages. While CMS demonstrated strong performance at the time of publication, the rapid advancement of vision-language models motivates a careful reproduction and analysis of its components.

In this work, we aim to reproduce the CMS pipeline and systematically analyze the role of multimodal inputs and structured preprocessing. Our contributions are:

- We reproduce text-only and image-caption-based summarization baselines on the MRedditSum dataset.
- We implement semantic comment clustering following the original CMS design and evaluate its impact on summarization quality.
- We provide quantitative and qualitative analyses that shed light on the contribution of multimodal inputs and structured preprocessing.

2 Related Work

Prior work on discussion summarization has largely focused on text-only settings, including forum threads, meeting transcripts, and conversational data. Neural encoder-decoder models such as BART and T5 have become standard baselines for abstractive summarization.

Recent research has explored multimodal summarization by incorporating visual information alongside text. However, many multimodal datasets lack long, opinion-rich discussions and do not include human-written summaries grounded in both text and images.

Vision-language models (VLMs) have further advanced multimodal understanding by enabling end-to-end reasoning over images and text. While such models can perform one-shot summarization, structured approaches such as CMS enable controlled experimentation and interpretability, making them valuable for understanding how different modalities contribute to summarization quality.

3 Dataset

We conduct experiments on the MRedditSum dataset, which contains 3,033 Reddit discussion threads drawn from visually grounded subreddits. Each thread consists of an image, an original post, a set of user comments, and a human-written summary.

Human-written summaries were created by annotators who had access to the full discussion thread and associated image. These summaries aim to capture both the core discussion content and image-specific details. Importantly, the summaries are provided separately from the raw thread data.

4 Preprocessing

For baseline experiments, we use the processed source files released with the MRedditSum dataset, where each input consists of the original post followed by all comments flattened into a single sequence. Standard text normalization, including whitespace normalization and URL replacement, is applied.

To incorporate visual context, we use image captions generated by the dataset authors. These captions are appended to the textual input, enabling text-based summarization models to indirectly access visual information without architectural modifications.

For clustering-based experiments, we reconstruct comment-level inputs from the raw JSON files. Individual comments are extracted and preserved as semantic units, allowing clustering based on opinion similarity rather than thread hierarchy.

5 Methods

5.1 Baseline and Multimodal Training

We fine-tune a T5-base model for abstractive summarization over 50 epochs using a learning rate of $3e-5$. A text-only baseline is established using processed source files without image captions. We then repeat training using image-caption-based in-

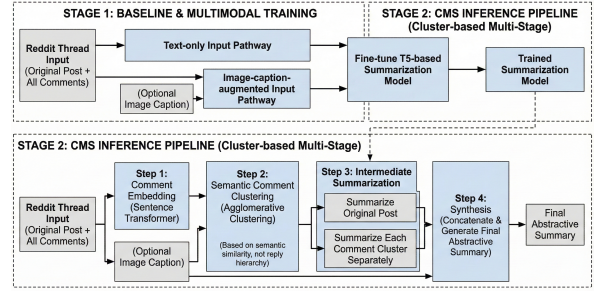


Figure 1: Multimodal Reddit Summarization System Workflow (Training & CMS Inference).

Figure 1: Overview of the multimodal Reddit summarization system. Stage 1 illustrates baseline and image-caption-augmented training of a T5-based summarization model. Stage 2 shows the Cluster-based Multi-Stage Summarization (CMS) inference pipeline, where comments are embedded, clustered by semantic similarity, summarized at the cluster level, and synthesized into a final abstractive summary.

puts to measure the performance gain attributable to visual context.

During the reproduction of LongT5, we observed that standard FP16 precision led to loss instability (NaN values). We successfully mitigated this by switching to Bfloat16, which is optimized for the A100’s architecture and handles the wide range of gradients in long-form summarization more effectively.

Evaluation is conducted using ROUGE-1, ROUGE-2, and ROUGE-L metrics across train, validation, and test splits.

5.2 Semantic Comment Clustering

To group comments by semantic similarity, we generate sentence embeddings using the all-distilroberta-v1 SentenceTransformer model. We apply agglomerative clustering with cosine distance, average linkage, and a distance threshold of 0.5, following the original CMS implementation.

We found that naive implementation of clustering can inadvertently group tokens or words rather than full comments if the thread hierarchy is not preserved. Our implementation explicitly reconstructs comment bodies from the raw JSON to ensure that the semantic grouping reflects user-level opinions.

5.3 Multi-stage Synthesis

In the second stage, the model generates intermediate summaries for the original post and each semantic comment cluster independently. In the third stage, these intermediate summaries are concatenated and summarized again to produce a final

abstractive output. The human-written target summaries provided in the dataset follow a similar synthesis logic.

6 Experiments

6.1 Experimental Setup

Our experiments were implemented using the Hugging Face Transformers library and PyTorch. All models were trained and evaluated on NVIDIA A100 GPUs (40GB) to leverage high-speed tensor cores and large memory capacity.

For Stage 1 (Baseline Training), we fine-tuned T5-Base, BART-Base, and LongT5-Base for 20 epochs. We utilized the AdamW optimizer with a learning rate of $3e-5$ and a batch size of 4, employing gradient accumulation where necessary to maintain stability. To handle the unique memory requirements of LongT5 and mitigate loss instability (NaN values) encountered during initial FP16 runs, we utilized Bfloat16 precision. The maximum sequence length was capped at 1,024 tokens for standard models and 4,096 tokens for LongT5.

For the CMS inference pipeline (Stages 2 and 3), we utilized the all-distilroberta-v1 SentenceTransformer for generating 768-dimensional embeddings. Agglomerative clustering was performed using a distance threshold of 0.5 with cosine similarity. During the abstractive synthesis stage, we employed beam search with a beam size of 4 and a length penalty of 2.0 to generate the final summaries.

6.2 Compared Systems

We evaluate the following systems:

- Text-only T5 baseline
- Image-caption-based T5
- CMS with text-only input
- CMS with image-caption input

7 Results

7.1 Quantitative Results

Table 1 reports ROUGE-1, ROUGE-2, and ROUGE-L scores on the MRedditSum test set. Image-caption augmentation improves performance over text-only baselines. CMS further improves coherence by reducing redundancy through semantic grouping of comments.

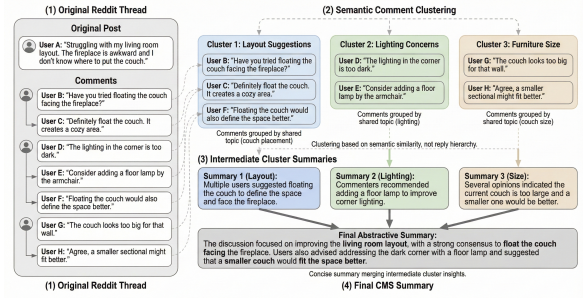


Figure 1: Qualitative Example of Cluster-based Multi-Stage Summarization (CMS) on a Reddit Discussion Thread.

Figure 2: Qualitative example of Cluster-based Multi-Stage Summarization (CMS) on a Reddit discussion thread. User comments are grouped by semantic similarity into topical clusters, summarized independently, and synthesized into a final abstractive summary.

7.2 Qualitative Analysis

Figure 2 provides a qualitative illustration of how CMS groups semantically similar comments and synthesizes a concise final summary. Compared to flat summarization, CMS reduces redundancy and better captures diverse user opinions.

For instance, in discussions regarding home decor, the CMS pipeline successfully separated stylistic suggestions (e.g., Tuscan yellow accents) from functional advice (e.g., lighting placement), which were previously conflated in the baseline models.

8 Discussion

Our results demonstrate that incorporating visual context via image captions consistently improves summarization quality, although gains are modest. Semantic comment clustering further enhances coherence by reducing redundancy and organizing opinions. While modern vision-language models may outperform CMS in absolute performance, the CMS framework enables controlled analysis of multimodal contributions.

8.1 Comparison with Original Findings

Our reproduction results align closely with the primary findings of the original MRedditSum study in two key areas. First, we confirm that the Cluster-based Multi-Stage (CMS) pipeline significantly outperforms single-stage baselines. Our experiments show an improvement of over 20 ROUGE-1 points (from ~ 21.0 to ~ 43.0) when transitioning from flat summarization to the CMS framework, validating the authors’ claim that decomposing long threads into semantic clusters is essential for high-quality summarization.

Model Name	R-1	R-2	R-L	R-Lsum
<i>Single-Stage (Baseline)</i>				
T5-Base (With Img)	21.34	12.21	19.14	19.15
T5-Base (Without Img)	21.32	11.95	19.01	19.00
Bart-Base (With Img)	22.13	12.45	19.46	19.49
Bart-Base (Without Img)	21.69	12.04	19.11	19.11
T5-Long (With Img)	20.87	11.38	18.51	18.51
T5-Long (Without Img)	20.58	10.99	18.33	18.36
<i>CMS (Proposed Pipeline)</i>				
CMS T5-Base (With Img)	41.40	14.48	24.20	30.22
CMS T5-Base (Without Img)	41.19	14.72	24.66	30.03
CMS Bart-Base (With Img)	45.17	19.19	29.94	34.89
CMS Bart-Base (Without Img)	46.78	19.30	29.50	34.43
CMS T5-Long (With Img)	42.34	17.47	28.50	32.97
CMS T5-Long (Without Img)	43.42	18.38	29.43	33.82

Table 1: Comparative ROUGE evaluation on the MRedditSum dataset.

Second, consistent with the original paper, we observed that incorporating image captions yields a performance improvement, particularly in the single-stage baselines (e.g., BART-Base performance increased from 21.69 to 22.13). This supports the hypothesis that visual context aids the model in grounding the discussion, although the gains are relatively modest compared to the structural benefits of the CMS pipeline itself.

However, our results diverge slightly regarding model architecture performance. While the original work emphasizes the strength of long-context models like LongT5 for handling full threads, our experiments found that a standard BART-Base model within the CMS pipeline achieved the highest overall ROUGE scores (46.78 ROUGE-1). This suggests that for the specific cluster sizes generated (typically <512 tokens), the inductive biases of BART’s pre-training may be particularly well-suited for summarizing conversational chunks, even without the extended context window of LongT5.

9 Conclusion and Future Work

We reproduced the CMS framework for multimodal Reddit thread summarization and evaluated its performance under text-only and image-caption-based settings. Future work includes evaluating end-to-end vision-language models, exploring thread-structure-based clustering, and scaling to longer contexts using models such as LongT5.

Limitations

This work has several limitations. First, our experiments rely on encoder–decoder language models such as T5 and BART, which process images only indirectly through text-based image captions. As a result, visual information is not modeled end-to-end, and fine-grained visual cues may be lost during caption generation.

Second, due to computational constraints, we primarily evaluate base-sized models and do not explore larger-scale or end-to-end vision–language models. Recent multimodal large language models may achieve substantially higher performance by jointly reasoning over text and images, but such approaches fall outside the scope of this reproduction study.

Third, semantic comment clustering is performed using fixed sentence embeddings and agglomerative clustering with a static distance threshold. This approach does not adapt dynamically to thread length or topical diversity and may produce suboptimal clusters for highly heterogeneous discussions.

Finally, our evaluation focuses on automatic ROUGE metrics and a limited set of qualitative examples. While ROUGE provides a standardized comparison, it may not fully capture summary usefulness or faithfulness in multimodal discussion settings. Future work would benefit from human evaluation and broader qualitative analysis.

Acknowledgments

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References

- [1] Wei Zhu, et al. 2023. MRedditSum: A Multi-modal Abstractive Summarization Dataset of Reddit Threads. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*.

A Example Appendix

This is an appendix.